

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY AND ENGINEERING
CHAMOS Matrusanstha Department of Mechanical Engineering

Computer Aided Manufacturing (ME 408.01)

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ME – 408.01 COMPUTER AIDED MANUFACTURING	
CO1	Understand the changes brought in the product cycles with the advent of CAM systems.
CO2	Understand emerging trends in CNC and Automation.
CO3	Apply their knowledge to prepare part program for machining processes.
CO4	Understand advance philosophy in the field of manufacturing.
CO5	Outline the working behind readily available Computer Aided Manufacturing software.

List of Experiment ME – 408.01 COMPUTER AIDED MANUFACTURING		
Sr. No.	Title	Course Outcomes
1	Construction and Working of NC/CNC Machine Tools	CO1
2	Configuration of CNC Machines	CO1, CO2
3	Manual Part Programming for CNC Milling & Lathe	CO3
4	Computer Assisted Part Programming	CO3
5	Programming For Machining Centre Using Cam Software	CO3, CO5
6	Study of tool paths for Milling and Turning cycles	CO3, CO5
7	Rapid Prototyping	CO4
8	FMS – Introduction & System Elements	CO4
9	Group Technology – Matrix Formation Algorithms	CO4
10	CIM – Emerging Technologies	CO4

List of Assignment ME – 408.01 COMPUTER AIDED MANUFACTURING		
Sr. No.	Title	Course Outcomes
1	Assignment 1- NC/CNC Machine Tools	CO1, CO2
2	Assignment 2- Part Programming	CO3, CO5
3	Assignment 3- RP, FMS, GT and CIM	CO4

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Date:

Practical No. 1
CONSTRUCTION AND WORKING OF NC/CNC MACHINE TOOLS

AIM

To study various parts of NC/CNC machine tools and their working

OBJECTIVES

To study different parts of CNC machine tools in general.

THEORY

It is important to know how your CNC machine is constructed. Understanding your machine's construction will help you to gauge the limits of what is possible with your machine. The CNC programmer understands the basic workings of the CNC machine in order to get the most from the CNC machine tool.

For a universal style slant bed turning center, for example, the programmer should know the most basic machine components, including bed, way system, headstock & spindle, turret construction, tailstock, and work holding device. Information regarding the machine's construction including assembly drawings is usually published right in the machine tool builder's manual. Truly, the more you know about your machine's capacity and construction, the easier it will be to get comfortable with the machine. Any CNC machine tool essentially consists of the following parts:

- a) Part program
- b) Program input device
- c) Machine Control Unit
- d) Drive System
- e) Machine Tool
- f) Feed Back System

QUESTIONS

1. What are the special constructional features of CNC machine tools?
2. Describe with sketch the working and construction of recirculating ball screw used in CNC machine tools stating advantages. What is preloading?
3. Describe with sketch an automatic tool changer (ATC).
4. Describe the principle of working of an automatic pallet changer (APC).
5. List the types of drives used for stepless control of NC machine tool spindles.

REFERENCES

1. Computer Aided Manufacturing by P N Rao, N K Tewari & T K Kundra
2. CAD/CAM/CIM by P. Radhakrishnan & S. Subranarayan
3. CNC Machines by P. Radhakrishnan

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Practical No. 2
CONFIGURATION OF CNC MACHINES

AIM

To study the configuration of CNC machines

OBJECTIVE

To study the constructional and control features of CNC Lathe & Milling

THEORY

CNC machine tools have evolved from the basic machines to the very sophisticated systems that provide lot of flexibility as well as productivity. Generally speaking, any CNC machine tool consists of the following units: Computers, Control systems, Drive motors & Tool changer.

The control systems used on NC/CNC machines are open-loop and closed-loop. The open-loop control system does not provide positioning feedback to the control unit, while in the closed-loop control this is monitored by the feed back device.

The majority of NC/CNC machine tools are equipped with automatic tool changers, such as magazines on machining centers & turrets on turning centers. They allow tool changing without the intervention of the operator.

QUESTIONS

1. How are the axes designated in NC/CNC machine tools? Sketch and designate the axes of CNC machine tools in the laboratory.
2. Explain with the sketch, the types of control with reference motion in NC/CNC machines.
3. Discuss difference between open-loop and close-loop systems.
4. List the typical specifications of CNC machines available in the laboratory.

REFERENCES

1. Computer Aided Manufacturing by P N Rao, N K Tewari & T K Kundra
2. CAD/CAM/CIM by P. Radhakrishnan & S. Subranarayan
3. CNC Machines by P. Radhakrishnan

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CNC LATHE SPECIFICATION

Machine Model: TITAN CNC LATHE

Capacity	Max. Turned Length Max. Turned Diameter Swing Over Bed	200 mm 100 mm 250 mm
Travel Range	Max. Travel • X – axis • Z - axis	125 mm 200 mm
Spindle	Spindle Bore Spindle Speed Range	50 mm 200 to 2500 RPM.
Feed Rate	Max. Feed Rate of X, Z axis Rapid Traverse Speed	1000 mm/min 1 to 1000 mm/min
Indexing Turret	Turret Type No. of Tool station Tool Shank • Square • Round	Linear Gang Type. 4 (Max.) Sq. 25 x 25 mm 30 mm dia. (Approx.)
Drive	Spindle Motor (H.P./KW) Axis Motor (Feedback device type) Axis Ball Screw	1 to 1.5 H.P. Servo motor (Incremental Encoder type) 20 mm dia.
Accuracy	0.02 mm	-
Repeatability	0.02 mm	-

Controller Specifications

- PC based 32 bit microprocessor technology
- Standard G & M codes, backlash compensation
- Cable of 3D simultaneous movement
- Tool path graphics
- Capability to store over 10,000 part programs
- Command type incremental & absolute.
- Data entry by key board, floppy drive
- Program memory good for 20,000 block program.

General Specifications

- Slide ways: Trucite (Teflon based) lined no metal-to-metal contact.
- Lubrication system: Central lube system with pump
- Machine weight: 325 kg

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CNC MILLING SPECIFICATION

Machine Model: TITAN CNC MILLING

Table	Work Area (L x B) T – Slot	420 x 125 mm 11 mm, 80 mm center dist.
Travel Range	Max. Travel Range • X – Axis • Y – Axis • Z – Axis Dist. from spindle nose to table surface Dist. from spindle center to column	190 mm 125 mm 200 mm 210 mm 210 mm
Spindle	Spindle Taper Spindle Speed Range	ISO 40 150 to 5000 RPM.
Feed Rate	Feed Rate of X, Y, Z Axis Rapid Traverse Speed	1000 mm/min. 1000 mm/min.
Tool Magazine	Tool Capacity Tool Selection Tool Shank	Single Manual ISO 40
Drive	Spindle Motor (H.P./KW) Axis Motor (Type) Axis Ball Screw	AC variable frequency Servo motor. X, Y, Z 16 mm dia.
Accuracy	0.028 mm	-
Input Resolution	0.001 mm	-

Controller Specifications

- PC based 32 bit microprocessor technology
- FANUC compatible
- Standard G & M codes, backlash compensation
- Cable of 3D simultaneous movement
- Tool path graphics
- Capability to store over 10,000 part programs
- Command type incremental & absolute
- Date entry by key board, floppy drive
- Program memory good for 20,000 block program.

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Practical No. 3

MANUAL PART PROGRAMMING FOR CNC MILLING & LATHE

AIM

To study the concept of NC/CNC manual part programming

OBJECTIVE

To write the part program for the given exercise on CNC Milling & Lathe

THEORY

CNC part program is a detailed list of instruction that needs to be executed by the machine control unit (MCU) to achieve the final component shape. The machining sequence needed to manufacture a given part is broken down into small elements and written in a specific format understood by MCU. The programming language needs to be studied to develop meaningful part programs.

The following steps are involved in the development of a part program and it's proving:

- Process planning
- Axes and Tool selection
- Cutting process parameters planning
- Job and tool setup planning
- Machining path planning
- Part program writing and proving

EXERCISES

1. Write a manual part program for profile end milling the component shown in Figure 1. Assume speed, feed and necessary data clearly. Use absolute mode and draw the tool path. The cutter diameter is 20mm. For the same component write the NC program using cutter radius compensation. Also simulate the same using cam software.
2. For the component shown in Figure 2 make a part program on a vertical axis-machining center. Clearly show the set point and axes on the sketch of the part. Show all necessary calculations. For the same component write the NC program using canned cycles. Also simulate the same using cam software.
3. Write manual part programs using absolute mode for turning components shown in Figure 3. Assume speed, feed and necessary data stating clearly. Also simulate the same using cam software.

REFERENCES

1. Numerical Control & CAM by Kundra, Rao & Tewari
2. CAD/CAM – Principles & Applications by P.N. Rao
3. CAD/CAM/CIM by P. Radhakrishnan, S. Subramanyam

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MANUAL PART PROGRAMMING FOR CNC MILLING & LATHE

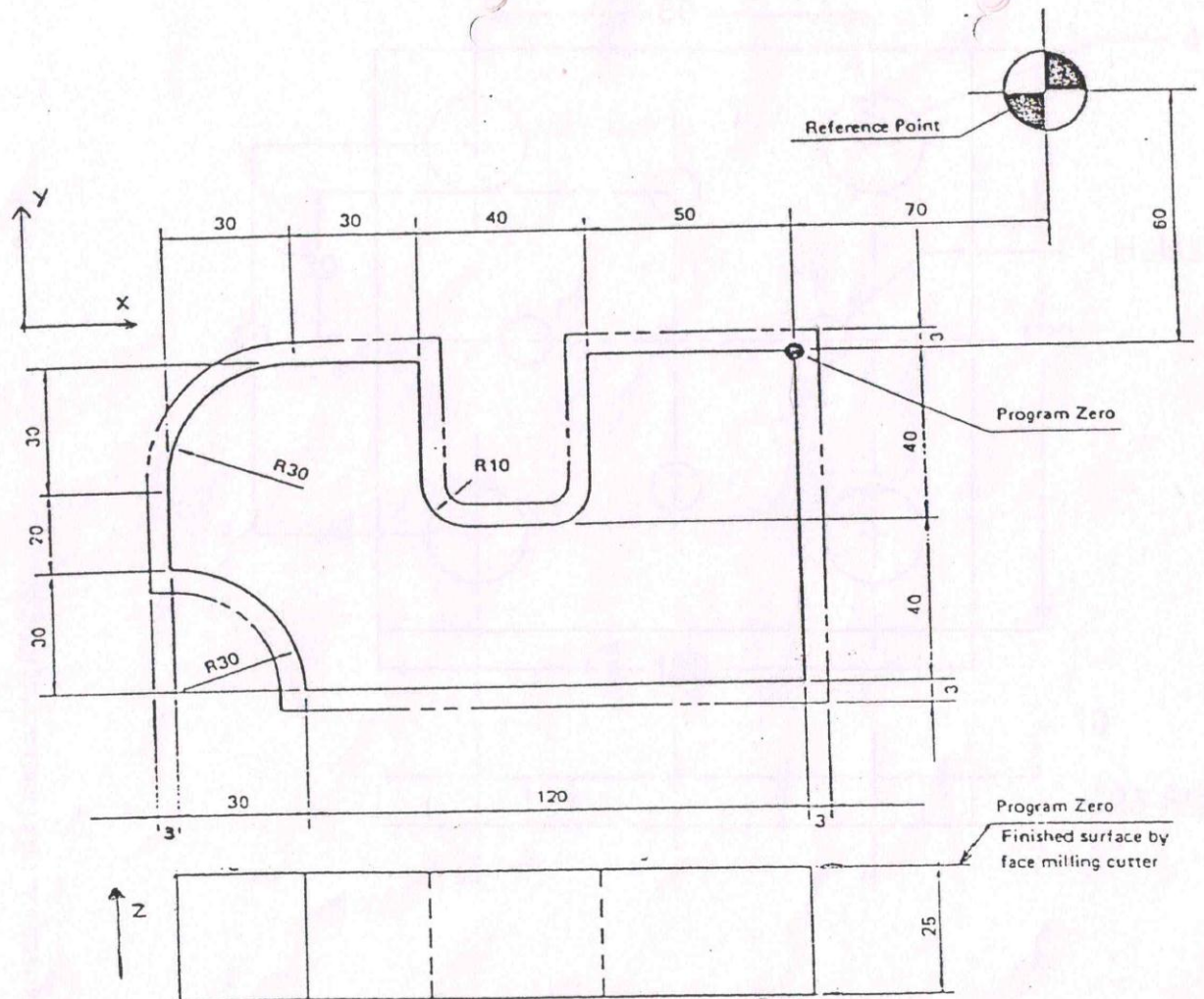


Figure 1

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Practical No. 3
MANUAL PART PROGRAMMING FOR CNC MILLING & LATHE

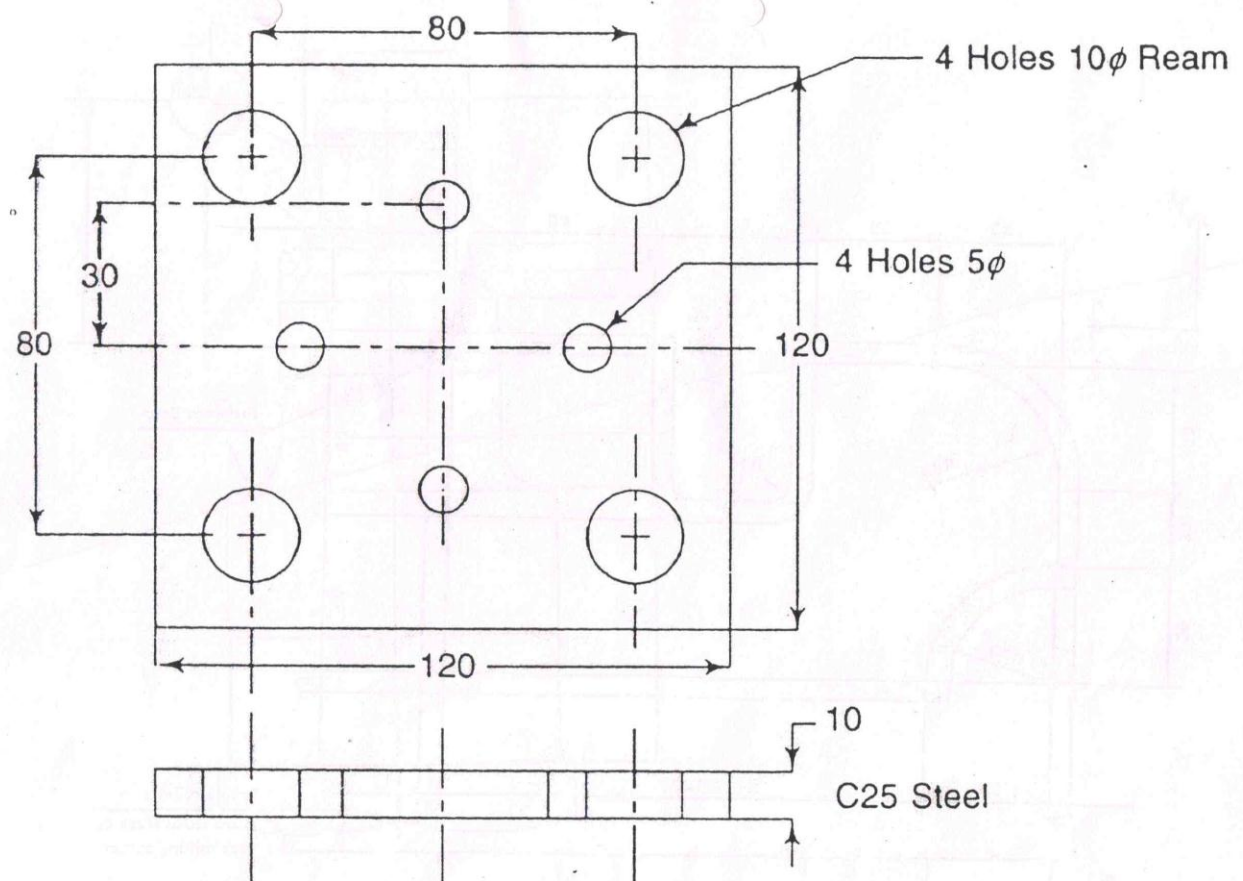


Figure 2

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MANUAL PART PROGRAMMING FOR CNC MILLING & LATHE

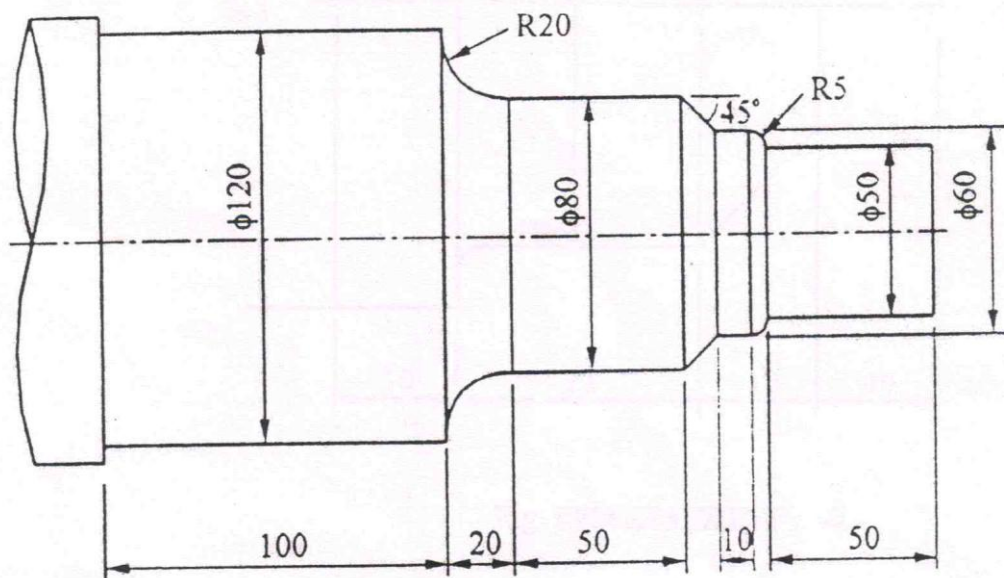


Figure 3

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Practical No. 4
COMPUTER ASSISTED PART PROGRAMMING

AIM

To study the Computer Assisted Part Programming

OBJECTIVE

To write the APT programming for the given exercise on Milling & Turning Center

THEORY

The APT (Automatically Programmed Tool) programming language was developed in early 1960s to assist engineers in defining the proper instructions and calculations for NC part programming. A great strength of APT is its ability to perform precise calculations for complicated tool paths when contouring on a three dimensional surface in a multi- axis programming mode.

The complete APT part program consists of the following four types of statements:

1. Geometry,
2. Motion,
3. Post Processor &
4. Compilation control.

EXERCISE

1. Write an APT program to machine the bracket shown in Figure 1. Assume the component to be 10 mm thick. The post processor statement is MACHIN/MMPOST, 3. The end mill used is 10 mm in diameter. Assume spindle speed as 1000 rpm and feed rate as 0.3 mm/rev.
2. Write a complete APT program to turn a shaft as shown in Figure 2. The post processor call statement is MACHIN/CLAT, 1. The tool has a nose radius of 3 mm. Assume spindle speed as 300 rpm and feed rate for machining as 0.3 mm/rev.

REFERENCES

1. Numerical Control & CAM by Kundra, Rao & Tewari
2. CAD/CAM – Principles & Applications by P.N. Rao
3. CAD/CAM/CIM by P. Radhakrishnan, S. Subramanyam

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Practical No. 4
COMPUTER ASSISTED PART PROGRAMMING

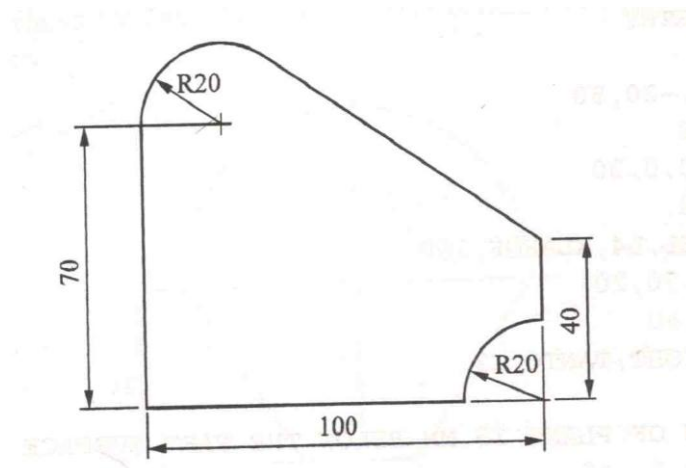


Figure 1

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Date:

Practical No. 4
COMPUTER ASSISTED PART PROGRAMMING

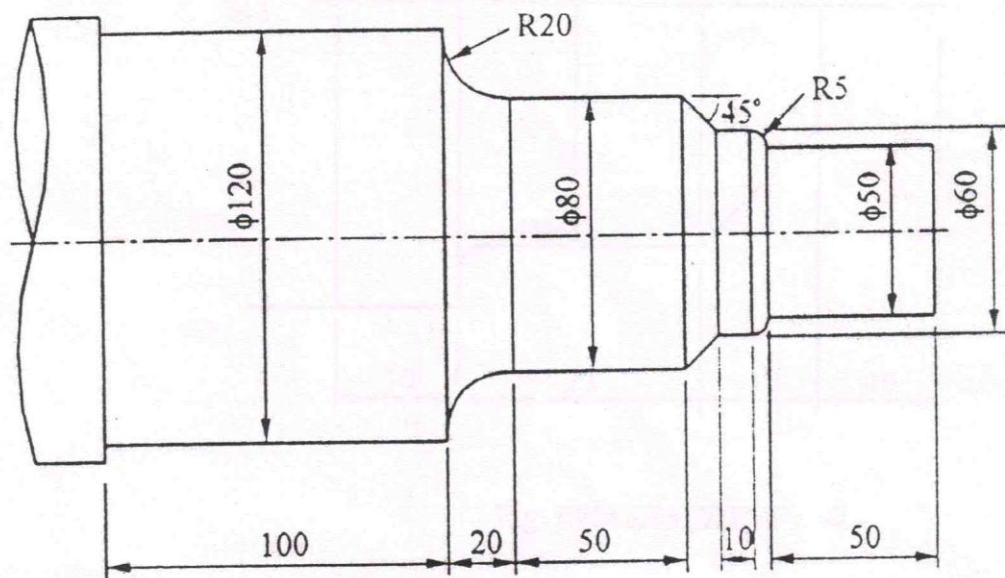


Figure 2

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Practical No. 5
PROGRAMMING FOR MACHINING CENTRE USING CAM
SOFTWARE

AIM

To refer the case study on CAD/CAM integration

OBJECTIVE

To understand the automatic part program generations for Machining & Turning Centre using Pro/Manufacturing.

THEORY

The integration of CAD and CAM has resulted in a paradigm shift in the methodology of NC programming. Today it is possible not only to generate CNC program required to manufacture a component directly from the CAD model but also design and model the fixturing set up, design the blank, process plan, select optimum process parameters, and simulate the machining operations on the CAD/CAM workstations to ensure that the program is capable of producing acceptable components. Today the geometry of the component can directly be obtained from the 3-D CAD model. CAM software can create the NC program from the geometry data.

There are several popular CAM packages available today. Some of them are listed below:

PRO/Manufacturing, CVCNC, I-DEAS generative Manufacturing, UG – Manufacturing, Surf CAM, Virtual Gibbs, Edge CAM, Prospector, Master CAM, Cimatron etc.

The steps involved in the part program development using CAM software are described in the next section. The starting point of CAM is the CAD file. A common approach is the program creation carried out using solid models or surface models. Data for program creation can also be obtained from SAT (ACIS solids), IGES, VDA, DXF, CADL, STL and ASCII file using suitable translators.

Procedure involved in NC program creation:

1. Create a manufacturing model from the design model and the work piece.
2. Set up the tool database. Tools must be defined before an operation is performed. Tool libraries can be created and retrieved for a manufacturing operation. Alternatively, they also can be created at the time of defining the manufacturing operation.

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3. Select the set up for the machining operation. A component may require more than one set up to complete the machining operation.
4. Fixtures are necessary at each set up. Fixtures can be displayed along with the work piece. This is to ensure that the path of the tools does not interfere with the tools. Standard fixture elements can be created as a library of parts and assembled before the manufacturing operation is commenced. These are also available as standard libraries.
5. Create a machinability database. Parameters like spindle speed and feed rate can be selected from the machinability database.
6. Create the manufacturing operations to generate the Cutter Location (CL) data. As each manufacturing operation takes place, material is removed from the work piece to simulate the actual manufacturing operation. This simulation is enhanced by the tool path being displayed for each operation.
7. If needed the CL data can be modified by modifying the operation parameters or by editing the CL data file.
8. Create a manufacturing route sheet at the end of the manufacturing session.
9. Post-process the CL file to create the NC program.

QUESTIONS

1. Develop the part program using CAM software for the pocket milling operation on a machining centre. Create your own geometric model and give the detailed procedure along with your assumptions.
2. Simulate the above part program on CNC milling & give comments on it.
3. Enlist and explain the activities & benefits of CAD/CAM integration in manufacturing.

REFERENCES

1. CAD/CAM/CIM by P. Radhakrishnan, S. Subramanyam & V Raju
2. Pro/Manufacturing Tutorial

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Date:

Practical No. 6
STUDY OF TOOLPATHS FOR MILLING AND TURNING

AIM

To study various tool paths during milling and turning operations.

THEORY

Manufacturing simulation software's like Master CAM provide various means of machining simulations for wide range of machining operations. In order to shape the stock to designed dimensions material is removed in various steps using a set of tools. The path traveled by the tool across the workpiece surface and its movement decides the cycle time for machining. It also helps to identify and eliminate the idle moves.

Here we will study such toolpaths available in Master CAM during Milling and Turning operations.

As we perform 2D and 3D machining, accordingly tool paths are available.

2D Tool paths in Mill – Contour, Drill, Pocket,

3D Tool paths in Mill – Roughing, Finishing, Clean-up Machining

Tool paths in Lathe – Roughing, Finishing, Threading and Grooving

Here types of various toolpaths are to be studied along with their use.

QUESTIONS

1. Classify all the toolpaths available in CAM Simulation software.
2. For exercises of practical no. 3 generate toolpaths and part program using CAM simulation software and compare the program with the manual part program stating conclusion.

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Date:

Practical No. 7 RAPID PROTOTYPING

AIM:

To study about rapid prototyping.

OBJECTIVE:

1. To understand the scope and utility of rapid prototyping.
2. To study various processes of rapid prototyping.

THEORY:

Rapid Prototyping is an automatic process of manufacturing physical objects directly from their CAD models without the use of any tools, dies, molds or fixtures specific to the geometry of the objects being produced.

Rapid Prototyping improves the agility of the companies to respond to these challenges. Rapid Prototyping offers total automation in converting the virtual models into physical ones. However, use of Rapid Prototyping is still limited to the manufacture of only prototypes, mostly of non-metallic materials

QUESTIONS:

1. Explain the principle of Rapid Prototyping its advantages and applications.
2. What is soft and hard prototyping?
3. Classify and explain Rapid Prototyping processes.
4. Explain five basic system elements that affect shape of the prototype obtained in any Rapid Prototyping process.

REFERENCES:

1. Rapid Prototyping: A Brief Introduction by A. Ghosh
2. Rapid Prototyping Technology: Selection and Application by Kenneth G. Cooper.
3. Rapid Prototyping: Principles and Applications by Chua Chee Kai, Leong Kah Fai, Lim Chu -Sing.

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Date:

Practical No. 8

FMS – INTRODUCTION & SYSTEM ELEMENTS

AIM

To study & discuss about system support equipments for FMS

THEORY

The concept of FMS was born in London in the 1960s when David Williamson, a research and development engineer, came up with both the name and the concept. At the time he was thinking of a Flexible Machine System, and it was in a machine shop that the first FMS was installed. This concept of decentralized computer control of machine tools, combined with idea of using machine tools for 24 Hrs per day, was the beginning of the FMS.

Principal benefits of FMS:

- 1 Inventory reduction.
- 2 Direct labor savings.
- 3 Increased asset utilization.
- 4 Floor space reduction.
- 5 Minimize delays, waiting for material movements.
- 6 Minimum material movements.
- 7 Control the bottlenecking of machine tools.

PROCESSING EQUIPMENTS

Turning Centers:

Increased and improved machine tool technology has taken the NC lathe beyond the scope of conventional turning operations. Approaching the flexibility of machining centers, a new array of NC turning center features and options has emerged that extends the turning center's capabilities far beyond its earlier predecessors.

NC lathes are classified in two types:

- 1 Vertical NC Turning Center : Modern adaptation of the manual Vertical Turret Lathes (VTLs)
- 2 Horizontal NC Turning Center.

Machining Centers:

Machining centers originated out of their capability to perform variety of machining operations on a work piece by changing their own cutting tools. Thus began with tool change system, later many other features and capabilities are added.

Machining just like turning centers, are classified as Vertical or Horizontal centers.

Vertical Machining Centers:

Widely accepted and used for flat parts where 3-axis machining is required on a single part face such as in mold and die work.

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Horizontal Machining Centers:

Widely accepted and used for boxy and heavy parts, because they lend themselves to easy and accessible pallet shuttle transfer when used in cell or FMS application.

QUALITY ASSURANCE EQUIPMENTS

Cleaning and De-burring Equipments:

These are less recognized member of the FMS as they perform the post- machining operations of little values.

It saves time and free employees to perform more meaningful work else here. Parts must be cleaned and de-burred before they can ever attempt to be accurately inspected, stocked or assembled.

Automated material movement and storage systems:

Material movement and storage with respect to cells and systems cover more than the traditional work piece flow and movement; they also include tool, fixture and pallet movement and storage to and from the processing stations and queue areas along with chip collection and removal.

Automated Guided Vehicles:

AGVs were first used in Europe after World War II as driverless tractors.

Types of guidance system for AGVs:

Tow Line, Wire Guided, Inertial Guidance, Infra-Red, Laser, Optical, Teach Type

Robot:

A robot is an automatic, servo-controlled, freely programmable, multipurpose manipulator with several areas, for handling of work piece, tools or special devices. Variably programmed operations make the execution of a multiplicity of tasks possible

Automated Storage and Retrieval Systems:

The aim of ASRS is to deliver the right material to the right place at right time. The concept of high rise, high density storage and retrieval was considered a radical change in the inventory management and control, rather than a revolutionary breakthrough.

ASRSs are strictly warehouse tools that track incoming material and components, store parts, tools and fixtures and retrieve them when needed.

Conveyor and Pallet Floation System:

Conveyor:

It contains a fixed path over which components travels from one point to other point in a production or process industries.

Types of Conveyor Systems:

Overhead mounted:

Floor mounted:

Further classified as chain, roller or belt driven.

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Queuing Carrousels and Automated Work Changers:

They are part parking lots that can hold 6, 8, 12 or more pallets or work pieces in various stages of completion.

Queuing carrousels hold parts in queue that:

1. Are scheduled to be processed and are waiting for an open machine in order to begin processing
2. Are at some stage of process completion and are waiting because a required machine is unavailable
3. Have completed processing and are waiting to be unloaded

Cutting Tools and Tool Management:

It is most cumbersome and difficult activity in FMS. Its scope includes getting right tool to right place at right time.

QUESTIONS

- 1 Describe with sketch working of AGVs used in FMS stating advantages, limitations and safety measures.
- 2 State the functions and benefits of Robot in FMS.
- 3 Describe types of AS/RS used in FMS. And how the success of AS/RS is measured?
- 4 What is Palletizing? Discuss its importance in FMS.
- 5 Describe conveyor and Pallet floating systems used in FMS.
- 6 Explain in brief Tool management in FMS

REFERENCES

1. Flexible Manufacturing Cells & System by Luggen
2. CAD/CAM – Principles & Applications by P.N. Rao
3. CAD/CAM/CIM by P. Radhakrishnan, S. Subramanyam

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Date:

Practical No. 9
GROUP TECHNOLOGY – MATRIX FORMATION
ALGORITHMS

AIM

To study the Group Technology

OBJECTIVES

1. To study different types of coding systems, coding structures and their applications.
2. To generate the codes using particular coding system for the given part.

THEORY

Definition of GT (Group Technology)

“A manufacturing philosophy in which similar parts are identified and grouped together to take advantage of their similarities in manufacturing and design.”

Similarities in Design and Manufacturing

Design Attributes

Part Configuration (round, prismatic), Dimensional envelope (ratio - length / diameter), Surface integrity, Material type, Raw material state

Manufacturing Attributes

Operations, Batch sizes, Machine and cutting tools, Processing times

Part Family

“A collection of parts which are similar either because they possess similarities in geometric shape and size, or because similar processing steps are used in their manufacture.”

- Part families are a central feature of group technology.
- There are always differences among parts in a family but the similarities are close enough that the parts can be grouped into the same family.

Coding and Classification of Parts

Coding: Assignment of a symbol (or a set of symbols) to represent information

Classification: A protocol that is used to separate a large group of objects into separate sub-groups.

Therefore, we first estimate (or survey) all possible different shapes we may be dealing with. Next, we identify some "features" that define something distinguishing about the shape of a part.

Coding structures

- (a) Hierarchical Structure (Tree Structure) (Monocode)
- (b) Chain Structure (Polycode)
- (c) Hybrid Structure

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Types of coding system

OPITZ Coding System

The OPITZ Coding system uses the following digit sequence 12345 6789 ABCD
Later, 4 more digits were added to the coding scheme, in order to increase the manufacturing information. These last four digits are also called supplementary digits.

MICLASS Coding System

MICLASS stands for “Metal Institute Classification and System”. It is the most popular commercial system available system in US

It consists of 2 sections. 1st section is of 12 digit code and 2nd is of 18 digit code.

DCLASS Coding System

DCLASS stands for “Design & Classification Information System” is used for educational & research purposes. Companies used this coding system for its prototype development. It is an 8 digit-code.

Clustering Algorithms

Single-Linkage Cluster Analysis (SLCA)

It is a hierarchical machine grouping method known as **single-linkage cluster analysis** using similarity coefficients between machines. The procedure is to construct a tree called a **dendogram**.

Rank order clustering (ROC)

ROC is a well-known clustering technique that attempts to create a block diagonal form by repeatedly reallocating the columns and rows of a machine/part matrix according to binary values.

Direct clustering analysis (DCA)

DCA rearranges the rows with the left-most positive cells to the top and the columns with the top-most positive cells to the left of the matrix. After several iterations, all the positive cells will form diagonal blocks from the top left corner to the bottom right corner.

Bond energy analysis (BEA)

The bond energy analysis algorithm attempts to identify and exhibit the interrelations within each cell and the associations among the clustered groups by means of total bond energy. A bond is claimed to exist between each pair of the neighboring rows and columns if they have positive cells in the machine/part matrix. BEA begins with an arbitrarily selected column (or row). It then places that column with the greatest contribution to the total bond energy beside the assigned column (row). It repeats the same procedure for all the columns and rows. The method is applicable to problems of any size because the BEA has nothing to do with calculating the binary values.

Applications of GT

1. Design
2. Machine Selection
3. Factory layout and work-flow planning
4. Fixture Design

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5. Automobile Manufacturing

QUESTIONS

1. State the need of a structure and explain the types of structure used in coding stating illustrations and applications. Explain the types of structure used in OPITZ, CODE and MICLASS coding systems.
2. Explain the clustering by Bond Energy algorithm for part family formation with the help of appropriate illustration.
3. Apply the Rank Order Clustering technique to the part-machine incidence matrix in the following table to identify logical part families and machine groups. Parts are identified by letters and machines are identified numerically.

Machines	Parts								
	A	B	C	D	E	F	G	H	I
1	1								1
2		1					1		
3			1		1			1	
4		1				1	1		
5			1					1	
6						1	1		
7	1			1					
8			1		1				

REFERENCES

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2. CAD/CAM by Groover & Zimmers

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FACULTY OF TECHNOLOGY AND ENGINEERING

CHAMOS Matrusanstha Department of Mechanical Engineering

Computer Aided Manufacturing (ME 408.01)

Date:

Practical No. 10 **CIM – EMERGING TECHNOLOGIES**

AIM

To study the various emerging technologies in CIM environment, its application and scope.

THEORY

The point is, we have the best machine tools we can get with today's technologies. Now, we are going to add something to them that will not try to make the machines better but will try to make the parts better.

The area where technology is emerging are electronic materials, surface transportation, system management, intelligent process equipments, expert systems, micro & nano manufacturing, vision systems, high performance materials, ceramics, composites, aeronautics etc. Several of these areas are directly or indirectly affect the CIM, its development. And they are undoubtedly helps the manufacturer to achieve CIM.

Expert Systems

It is a branch of computer based artificial intelligence. It provides the services of the experts at shop floor in the absence of the human experts. The system may be interfaced to the end users or to an array of sensors and effectors to communicate with the plants and processes.

Advantages

- 1 Captures the expertise of the employees who may not be there tomorrow due to job change, retirement or death
- 2 Synergetic effect with the knowledge of several experts
- 3 Resulting decisions are consistent
- 4 Knowledge can be updated, revised and improved
- 5 Knowledge can be shared and used when an expert is busy or not available.

Computer vision

It is called as machine vision. It is defined as a system for automatic acquisition and analysis of the image to obtain desired data for interpreting or controlling the activities. In a broader sense, the term is applied to a wide range of the non-contact electro-optical sensing techniques from simple triangulation, profiling to a 3D object recognition technique.

QUESTIONS

- 1 Describe the process for implementing CIMS stating an illustration.
- 2 Discuss the application of computer vision in CIMS.
- 3 Discuss need and importance of shop floor data collection system. What are their functions?
- 4 Explain AI based scheduling in CIM environment.
- 5 Discuss the area where expert systems are applicable.

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